

SAMUEL GIRMA

Database Specialist | PostgreSQL · TypeORM · Prisma · Redis
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SUMMARY

Final-year Computer Science student and Software Development Intern at Vintage Technologies PLC, working daily on the PostgreSQL, TypeORM and Redis data layer of Lumina Academy, a production multi-role Learning Management System. Experienced in relational data modeling, permission and workflow data design, Redis caching and cache invalidation, vector search with pgvector, and verifying data correctness through hands-on QA.

EXPERIENCE

Software Development Intern

Jul 2026 – Present

Vintage Technologies PLC, Addis Ababa, Ethiopia

- Work on the data layer of **Lumina Academy**, a multi-role LMS for students, instructors and admins, built on a NestJS 10 REST API with **PostgreSQL, TypeORM and Redis**.
- Extended access control from a static role matrix to **versioned, account-level permission overrides** with explicit-deny rules and a 4-way permission diff, adding **Redis cache invalidation** so cached permissions stay consistent with stored data.
- Built data-driven workflows that depend on consistent records: quiz and assignment retry overrides, waivers, certificate issuance with public verification, and instructor payout approval flows.
- Built analytics dashboards (Recharts) over platform data and real-time notifications with Socket.IO.
- Hardened authentication with session lifecycle handling, JWT cookie expiry, forced password change for new and reset accounts, and a unified exception filter; fixed a guard bug so permission checks covered every module.
- Implemented secure AES-128 HLS video streaming with token authorization and a protected PDF viewer that blocks content access through shared links.
- Performed manual QA on the **Quick Delivery** logistics platform: owned the Field Operations track of the admin dashboard (live orders, trips, riders, businesses, emergency alerts), executed a 76-case business-portal test pack, and tested the customer and driver mobile apps for role-based access, OTP login and validation. Reported bugs with screenshots, reproduction steps and severity ratings before production sign-off.

Freelance Web Developer

2023 – Present

Self-employed, Addis Ababa, Ethiopia

- Designed, built and deployed 3 client websites end-to-end, from requirements and UI design to development and launch.

PROJECTS

HUKM – Ethiopian Criminal Law Analysis Tool

hukm-et.vercel.app

- Architected a RAG pipeline over the Ethiopian Criminal Code (Proc. 414/2004) using **Supabase with pgvector** for vector-embedding retrieval and a 7-step LLM reasoning process, returning source-cited answers with multi-turn chat.

Stack: Next.js 14, TypeScript, Tailwind CSS, Supabase, pgvector, NVIDIA NIM API

StudyLens – AI-Powered Learning Platform

study-lens.vercel.app

- Built a learning platform on **PostgreSQL and Prisma** with AI-powered search, FSRS spaced-repetition scheduling and personalized analytics dashboards.

Stack: Next.js 15, TypeScript, PostgreSQL, Prisma, NextAuth.js, Zustand, TanStack Query

CV Builder Platform

cv-builder-platform.onrender.com

- Built a full-stack CV platform on **PostgreSQL** with an AI-driven ATS scanner, automated cover-letter generation and multi-template PDF export.

Stack: Node.js, Express.js, PostgreSQL, JWT, PDFKit, NVIDIA API

TECHNICAL SKILLS

Databases: PostgreSQL, MySQL, SQL, Redis, Supabase, pgvector (vector embeddings)

ORMs & Data Access: TypeORM, Prisma, REST APIs, GraphQL

Backend: NestJS, Node.js, Express.js, JWT, Socket.IO

Languages: SQL, TypeScript, JavaScript, Python, Java, C++, PHP

Frontend: React, Next.js, Tailwind CSS, shadcn/ui, Zustand, TanStack Query, Recharts

Testing & QA: Manual and Functional Testing, Test Case Execution, RBAC Testing, Mobile App Testing, Bug Reporting

Tools: Git, GitHub, Docker, VS Code, Figma

EDUCATION

BSc in Computer Science

2022 – Expected Jul 2027

American College of Technology, Addis Ababa, Ethiopia

Relevant coursework: Databases, Data Structures, Algorithms, Computer Networks, Software Engineering

CERTIFICATIONS & LANGUAGES

Certification: Huawei Computer Networks – CRA Training Program, Huawei, 2025

Languages: English (Fluent), Amharic (Native), Arabic (Basic)